

CHALLENGE TITLE

Power, Seen From Orbit

How might we predict what a wind or solar park will really produce over its 30-year life, in a climate that no longer matches the past?

PARTNER

EnviroTrust helps energy producers (such as municipal utilities) and grid operators minimize the financial and operational risks tied to climate change and extreme weather.

THE PROBLEM

A wind or solar park is built to run for 30 to 40 years. Before it goes up, someone has to estimate how much energy it will produce over that whole life. That single number drives what happens next: whether the project gets financed, at what rate, what it can be insured for, and what it sells for later. For a park already running, the same number keeps getting re-asked every time it is refinanced, reinsured, or sold, and now there is a real track record to check it against.

Almost all of that estimate comes from the past. The standard method builds a “typical year” out of one or two decades of historical weather and projects it forward, assuming the next thirty years look like the last twenty. They won’t. Wind patterns are shifting, heat cuts what panels and electronics can put out, and storms and heatwaves that used to be rare are showing up more often. The forecast describes a climate that isn’t there anymore.

That error is expensive, and it builds up. Being off by a few percent a year, every year for thirty years, is the difference between a park that covers its loan and one that doesn’t. Operators see it as revenue that never shows up. Lenders see loans against assets that underperform, and insurers see claims they didn’t price for. The layout of the park barely makes it into the standard forecast either, even though wind drops off behind each turbine and heat pools across a solar field.

The data to fix this already exists: satellite imagery, forward-looking climate models, public registries of every installed park, and EnviroTrust’s climate API. Put them together and they can describe how one specific park, in its real location and layout, will actually perform over its life. Almost nobody has connected them into a number an investor would put their name to. That’s what you’re building.

THE OPPORTUNITY

Most new power capacity going up today is wind and solar, and the gap between forecast and reality is turning into the biggest financial risk attached to it. A model that predicts lifetime output under a shifting climate is useful to almost anyone with a stake in that number: owners weighing what a park is worth, lenders and investors checking a deal, insurers setting a price, operators planning revenue, and people further from the deal table who are still tied to the result for decades. It is basically a stress test for energy assets, the physical version of the financial models these projects already run.

A few places this bites, any one a fair starting point:

- **Revaluing a park already running.** Its forecast was set years ago on a climate that has since moved. What is the rest of its life really worth now, and how far has reality drifted from the projection?
- **Due diligence on a deal.** The seller’s projection assumes a stable climate. What does the number look like once you drop that assumption?

- **Pricing an insurance policy.** Which parks carry heat and storm exposure that the historical record doesn't show?
- **Planning operations.** How does output drift year to year, and when does repowering or extra maintenance pay for itself?
- **Or your own.** The user doesn't have to be on this list, and the strongest entries often aren't.

CONSTRAINTS (hard boundaries only)

- Model a real park that is already operating. Use real locations, real layouts, and public asset data. Climate and weather inputs may be modeled or scenario-based; the site and the hardware may not be invented.
- Every prediction carries its uncertainty. A single lifetime figure with no range, no scenario, and no stated confidence is not a prediction in this domain.
- A number that moves money must be explainable. Anything a lender, investor, or insurer would rely on has to trace back to its data and its assumptions.

DELIVERABLES

Standard LAUNCH Build Days deliverables, adapted for a prediction challenge:

1. **Solution concept with a clear value proposition:** what you are building, who it is for (an owner, lender, insurer, operator, or a user you choose), and why it beats today's history-based forecasting. No customer interviews or validation are expected for this challenge: name your user and make the case.
2. **A working prediction on real data:** a model or tool that takes at least one real park and produces its lifetime performance (or the gap between forecast and reality), with the climate dimension visible, for example its output under different climate scenarios. Modeled or scenario climate inputs are expected.
3. An Expo pitch: five minutes to present and demo, then four minutes of jury Q&A. Show the working thing, not slides.

EVALUATION CRITERIA

Standard LAUNCH Build Days criteria, with problem validation set aside for this challenge.

Criterion	Description
Solution Quality	Original and well-reasoned, with a clear user, a clear value proposition, and sound assumptions behind the model.
Prototype Tangibility	A model that actually runs on real park data and produces a credible lifetime prediction, not slides. A narrow, deep slice, one park and one climate dimension done well, beats a broad and shallow demo.
Business Potential	Could this become a real venture? Is there a market and a model beyond a single park or a single customer?
Pitch and Team	Clear, compelling, and confident. The team understands both what the prediction is worth and what the data can and cannot say.

Partner bonus (EnviroTrust): the best use of Copernicus climate data. Did the team bring the forward-looking projections from the Copernicus Climate Data Store (in Resources) into the prediction so the climate dimension actually moves the result, rather than leaning on historical weather alone?

RESOURCES

Start here - EnviroTrust climate API, api.envirotrust.eu/docs: forward-looking climate data, free, with access keys (one for each team) at kickoff. - Microsoft Global Renewables Watch, github.com/microsoft/global-renewables-watch: satellite-derived footprints of wind and solar parks worldwide. Your park geometry.

Climate and weather - Copernicus Climate Data Store, cds.climate.copernicus.eu: raw climate projections and reanalysis. Harder to work with than a packaged feed, and the data the partner bonus is built around. - Open-Meteo, open-meteo.com: free historical and real-time weather APIs, no key required.

Assets and prices - Marktstammdatenregister, marktstammdatenregister.de: Germany's official registry of wind and solar installations. Real parks with real specifications. - SMARD, smard.de, and the ENTSO-E Transparency Platform: electricity market and price data. What the energy is actually worth. ENTSO-E needs an API key you request in advance, so request it early; SMARD works without one.

Provided by EnviroTrust - Sample wind-turbine power curves, shared at kickoff: how wind speed becomes power for a real turbine.

Going further - Earth Observation VLMs, github.com/shuyansy/Earth-Observation-VLMs: vision-language models for satellite and earth-observation imagery, if you want to pull more out of the satellite layer. Suggested by EnviroTrust.